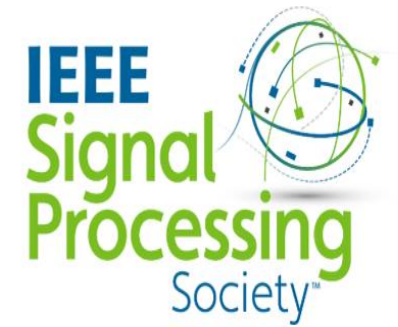




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# CXR-LT 2026: Multi-Label Long-Tailed Classification on Chest X-Rays

MIMIC-CXR-JPG-2.0.0



Postero-anterior :  
64470



Anterior-postero :  
100024



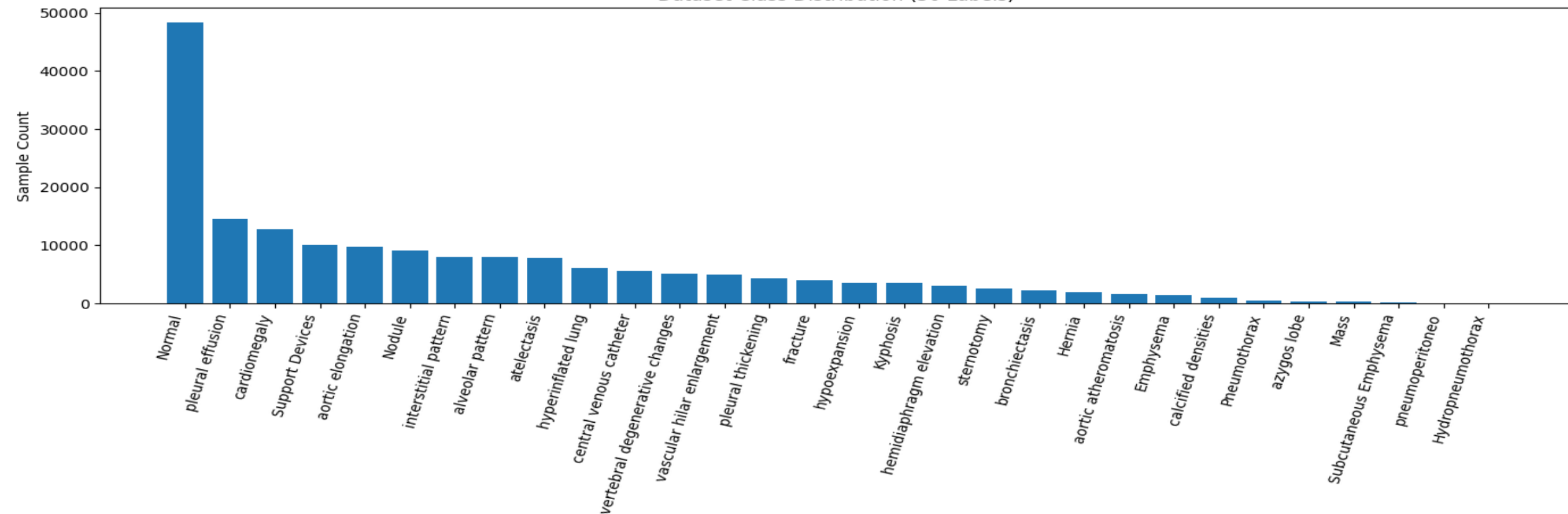
Left-lateral :  
24080



Lateral :  
57546

# CXR-LT 2026: Multi-Label Long-Tailed Classification on Chest X-Rays

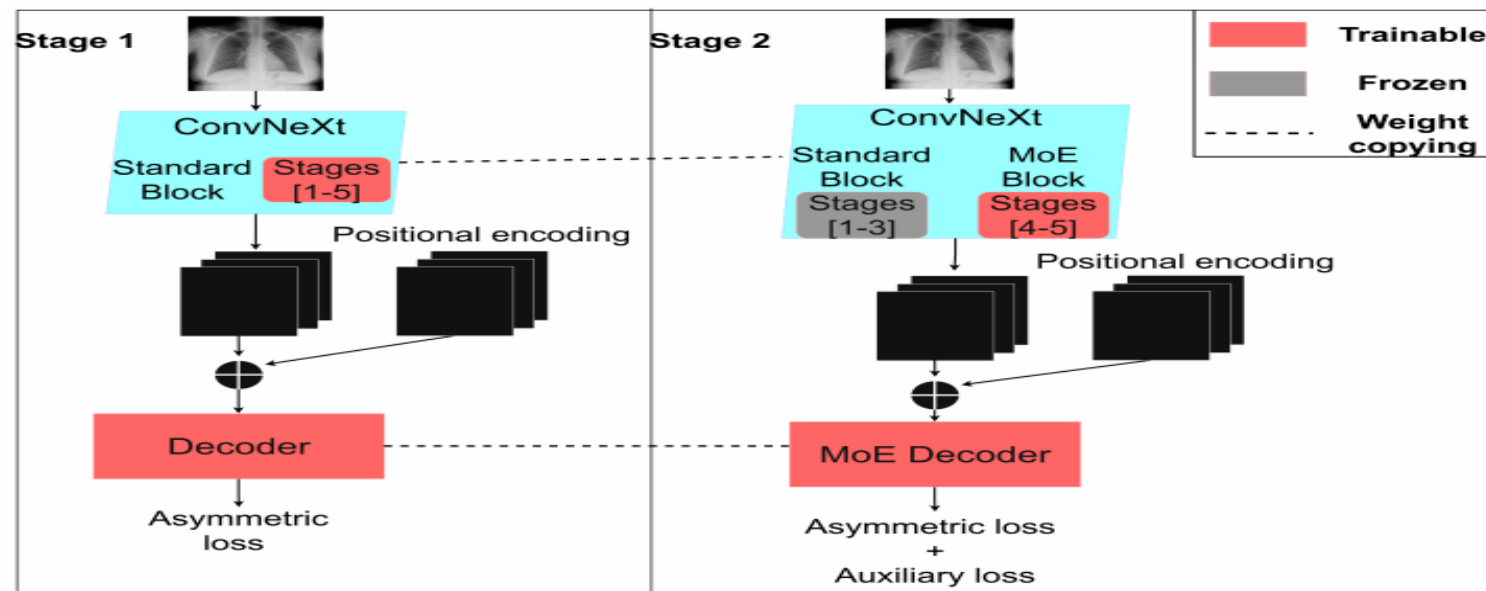
Dataset Class Distribution (30 Labels)



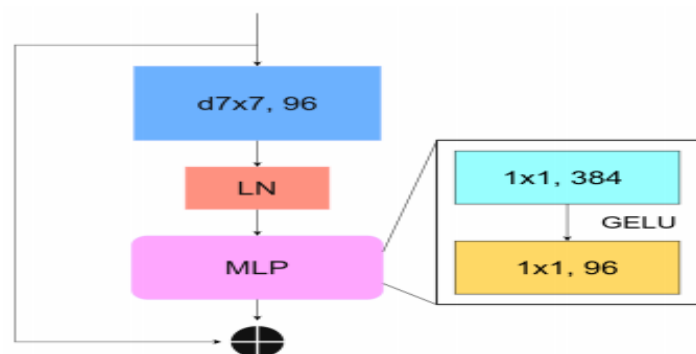
# Challenges

1. Label co-occurrence
  - ⇒ The tail disease often co-exist with head diseases which makes the learning the individualized signatures of the tail cases challenging
2. Imbalanced data distribution
  - ⇒ Inter-class: caused by long-tail data distribution
  - ⇒ Intra-class: negative classes dominate the multi-label classification, model is penalized heavily for false positives

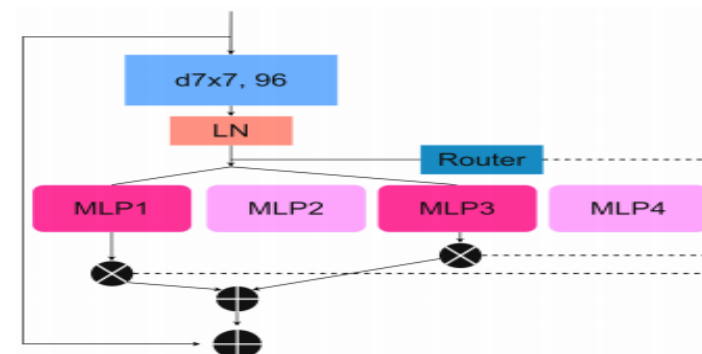
# Our Old Proposed Framework



(a) Two-stages of the framework

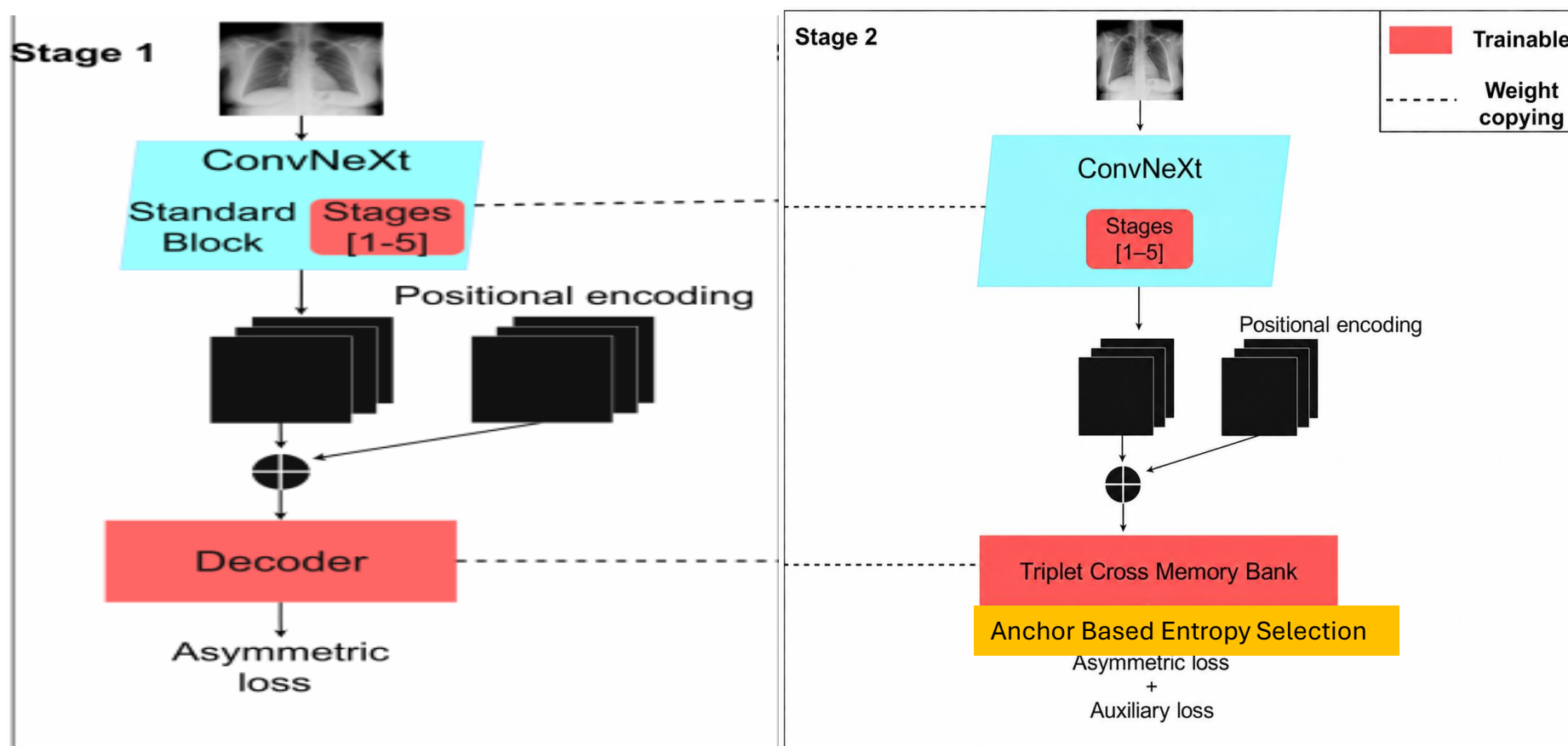


(b) Standard Block



(c) MoE Block

# Our New Proposed Framework (Under developping)

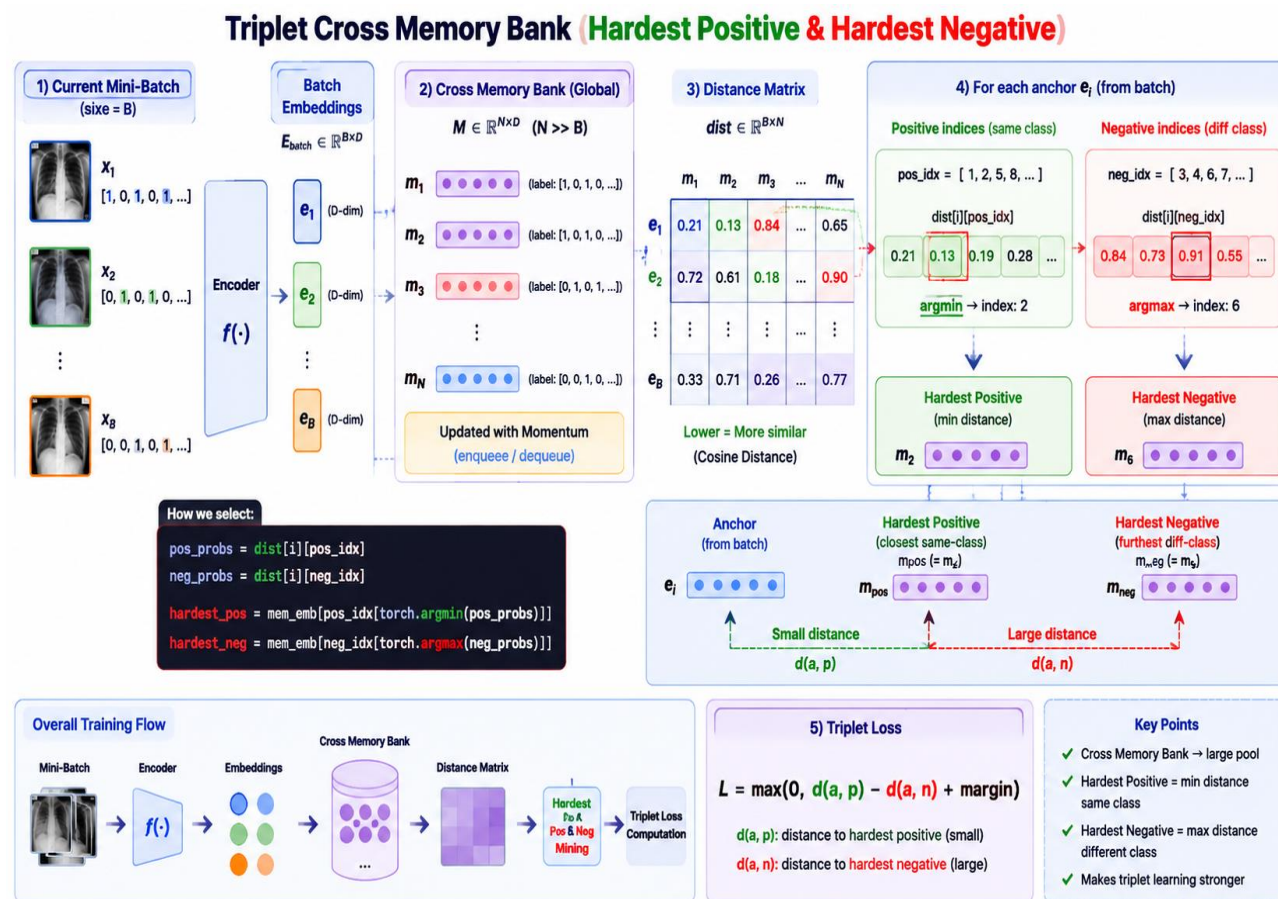
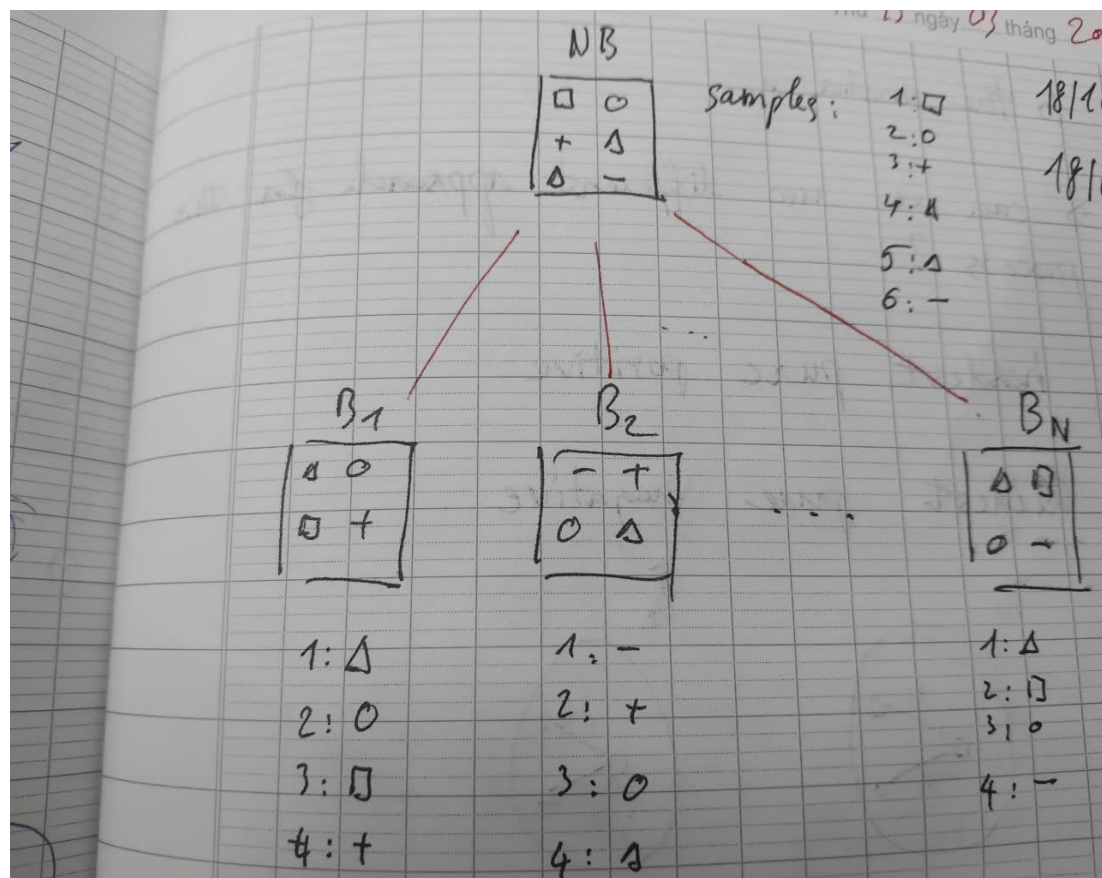




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# Triplet Cross Memory Bank Loss



## Triplet Cross Memory Bank + Anchor-Based-Entropy Selection

Why should integrate Anchor-Based-Entropy Selection?

How good the created Triplet? How good the anchor has selected?

Shannon Entropy Equation:

$$x_j^* = \arg \max \left( - \sum_i P_{\theta}(y_i|x) \log P_{\theta}(y_i|x) \right)$$

The anchor has selected by highest Shannon Entropy

## Experimental Setup

| Train and Val sets splitting | Sample size |
|------------------------------|-------------|
| Training set                 | 142000      |
| Internal Val set             | 17000       |

## Performance on internal development set

Image Size = 384

| Stage | mAP   | mF1   | mAUC  | mECE  |
|-------|-------|-------|-------|-------|
| 1     | 0.385 | 0.395 | 0.875 | 0.134 |
| 2     | 0.392 | 0.373 | 0.881 | N/A   |

Image Size = 512

| Stage | mAP   | mF1   | mAUC  | mECE  |
|-------|-------|-------|-------|-------|
| 1     | 0.385 | 0.395 | 0.875 | 0.134 |
| 2     | 0.405 | 0.395 | 0.884 | N/A   |

# Thank you for your attention